

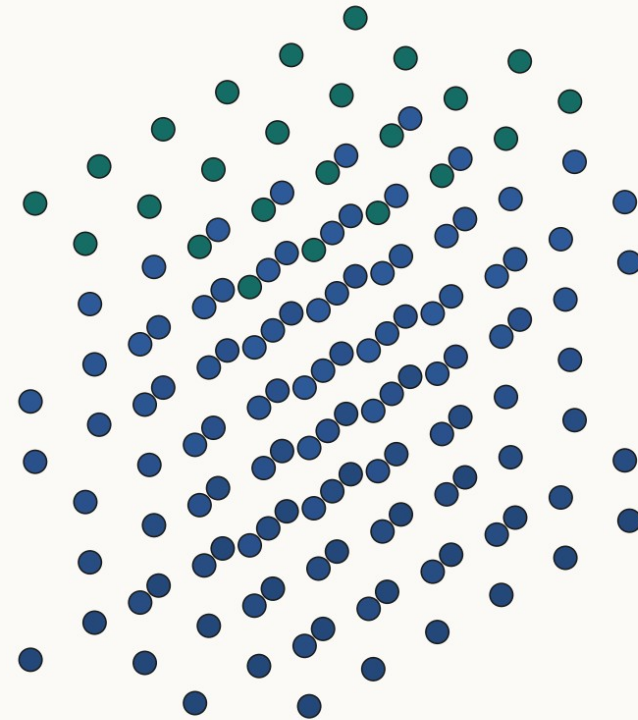
Interface Force Field

From atoms to a testable material
prediction

A visual guide to structure, simulation
and experimental evidence

Sean Florez · 3 October 2026

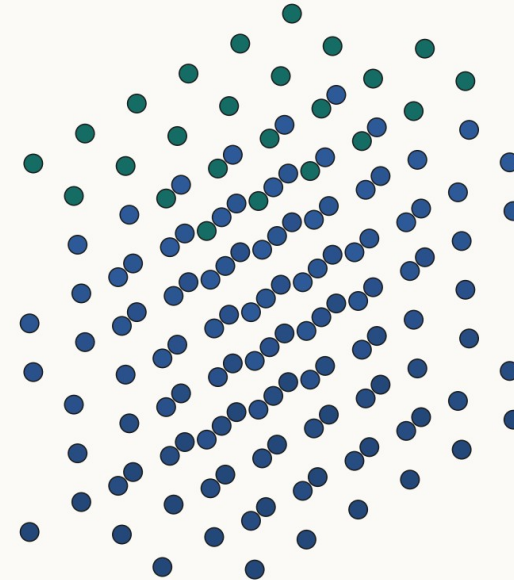
RHODIUM · A SOURCE-GROUNDED WALKTHROUGH



Begin with the decision

What would change if the predicted interface were wrong?

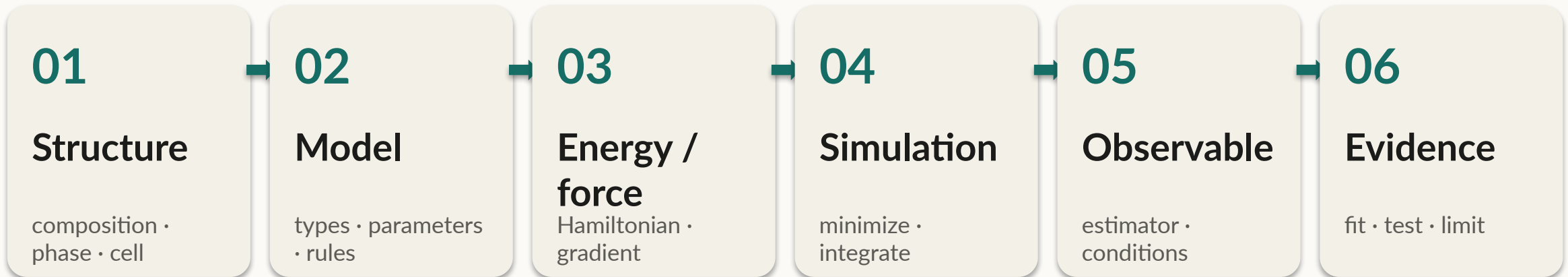
- | Question: what does it cost to expose a Rh surface?
- | Observable: (111) surface energy, γ_{111} , in J/m².
- | Scope: pure fcc Rh, a defined face and thermodynamic state.



DEFINE THE OBSERVABLE BEFORE CHOOSING THE METHOD

Keep every link in the chain visible

A structure is an input. A validated claim is an outcome.

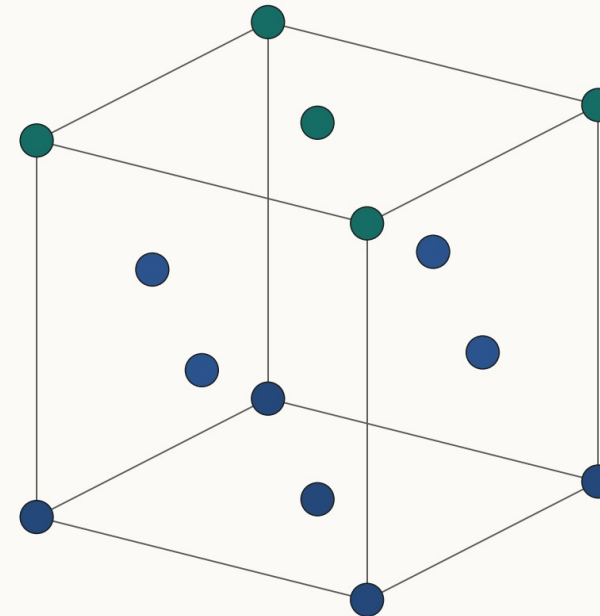


Separate the input, model, calculation and evidence for every claim.

Start with a real periodic crystal

Rhodium: fcc, Fm-3m, $a = 3.8032 \text{ \AA}$

- Corrected supplementary CAR: one symmetry-unique Rh site.
- Expand Fm-3m symmetry $\rightarrow$ 4 atoms per conventional cell.
- Render equivalent boundary sites to show the cell clearly.

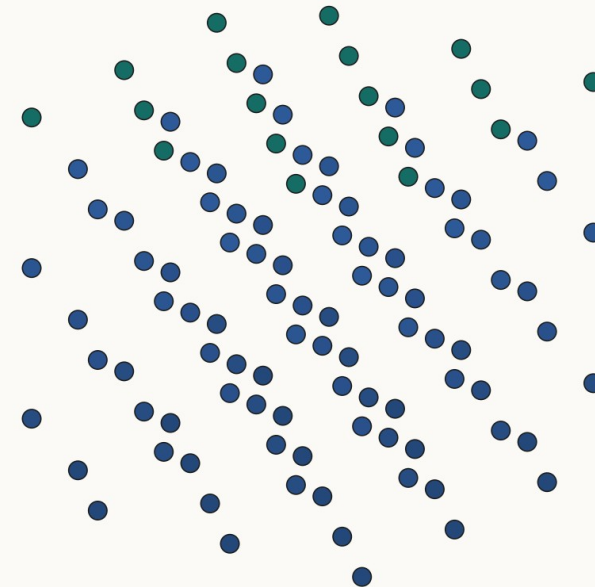


SOURCE GEOMETRY · STATIC IDEAL CELL

A supercell changes scale, not chemistry

Replicate the cell while preserving periodic meaning.

- 3 × 3 × 3 conventional cells → 108 Rh atoms.
- Bulk periodic boundaries reconnect opposing faces.
- Paper lattice calculations used 5 × 5 × 5 cells, not this display.



DISPLAY SUPERCELL · NO NEW SIMULATION

A surface introduces a physical boundary

Cut the crystal along a named plane.

- | Rh(111): a triangular in-plane neighbor arrangement.
- | Constructed display slab: $6 \times 4 \times 6$, 144 atoms.
- | State orientation, termination, vacuum and surface area.

Rhodium · from cell to surface

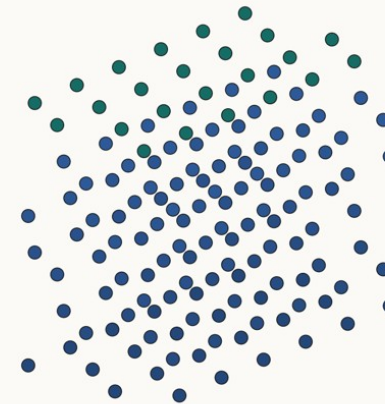
Expose a (111) surface | $a = 3.8032 \text{ \AA}$

(111) slab

144 atoms

Rh · bulk

Rh · top layer



$a = 3.8032 \text{ \AA}$

cubic cell parameter

Static ideal geometry from corrected Kanhaiya 2021 data; camera motion only

Animation is included in the media companion.

(111) · IDEAL SLAB · TOP LAYER IN TEAL

Coordinates do not define the model

The same atomic positions can produce different predictions.

- | | | |
|---|----------------------------|--|
| 1 | Geometry | species · coordinates · periodic cell |
| 2 | Chemical assignment | atom type · charge · bonded topology |
| 3 | Parameter package | functional form · coefficients · mixing |
| 4 | Numerical settings | cutoff · electrostatics · units · engine |

IFF is classical molecular mechanics

Its role is a chemically interpretable, compatible model.

Classical models

Pair potentials, many-body models, bonded and reactive families

IFF

Materials and interfaces within specific supported conventions

Sum only the applicable interactions

$U(R; \theta)$ is the model, with coordinates R and parameters θ .

Bonded

bonds · angles · dihedrals,
when applicable

Electrostatic

charges and their long-
range convention

Nonbonded

repulsion / attraction,
such as Lennard–Jones

PURE RH EXAMPLE: NEUTRAL ATOMS + LJ

Two parameters have physical meaning

Rh: set the length scale and the well depth.

12-6 LJ

$$R_{\min} = 2.757 \text{ \AA}$$
$$\epsilon = 7.84 \text{ kcal/mol}$$

9-6 LJ

$$R_{\min} = 2.807 \text{ \AA}$$
$$\epsilon = 6.38 \text{ kcal/mol}$$

$R_{\min}$ governs distance · ϵ governs well depth

TABLE 1 · EACH FAMILY IS A DISTINCT MODEL

Check the engine's parameter convention

The word “sigma” can denote different distances.

Paper convention

$$U = \varepsilon[(R_{\min}/r)^{12} - 2(R_{\min}/r)^6]$$

LAMMPS lj/cut

$$U = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$$

Exact conversion

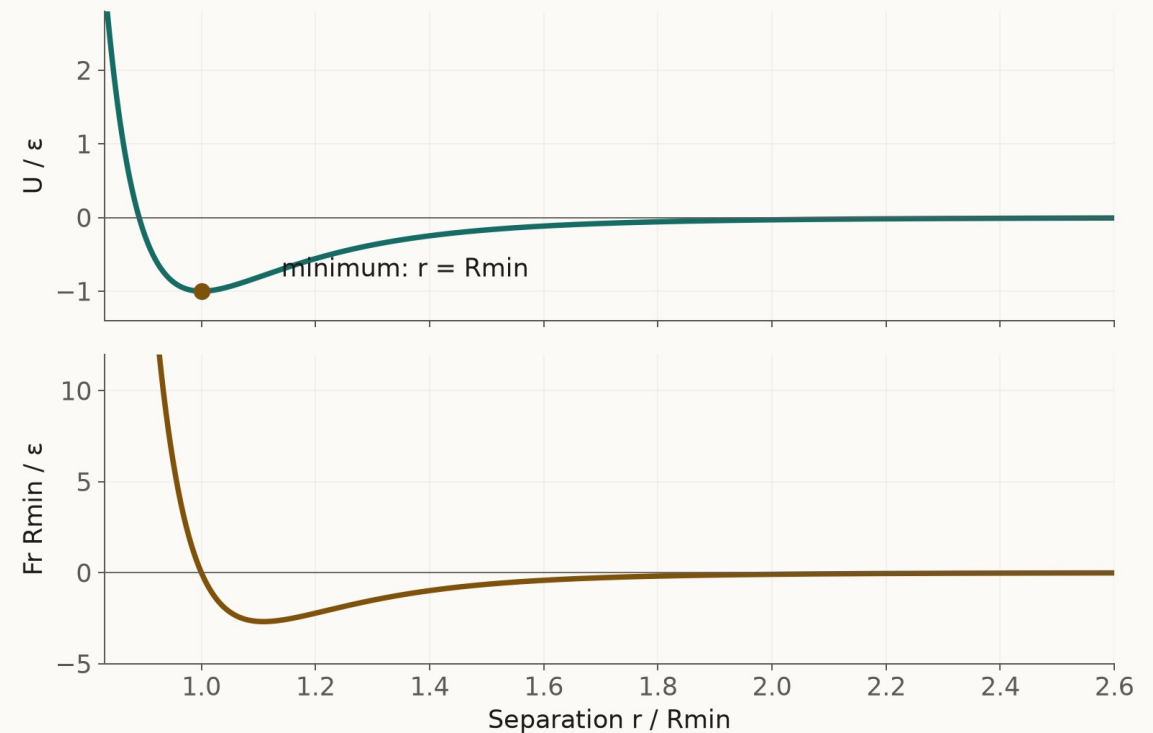
$$\sigma = R_{\min} / 2^{(1/6)} \cdot \varepsilon \text{ unchanged}$$

RH 12-6: $R_{\min} 2.757 \text{ \AA} \rightarrow \sigma \approx 2.4562 \text{ \AA}$

Forces are the slope of energy

$$F_i = -\nabla_{r_i} U(R; \theta)$$

- Too close: short-range repulsion dominates.
- Past the minimum: attraction points back inward.
- A pair minimum does not equal the bulk lattice constant.

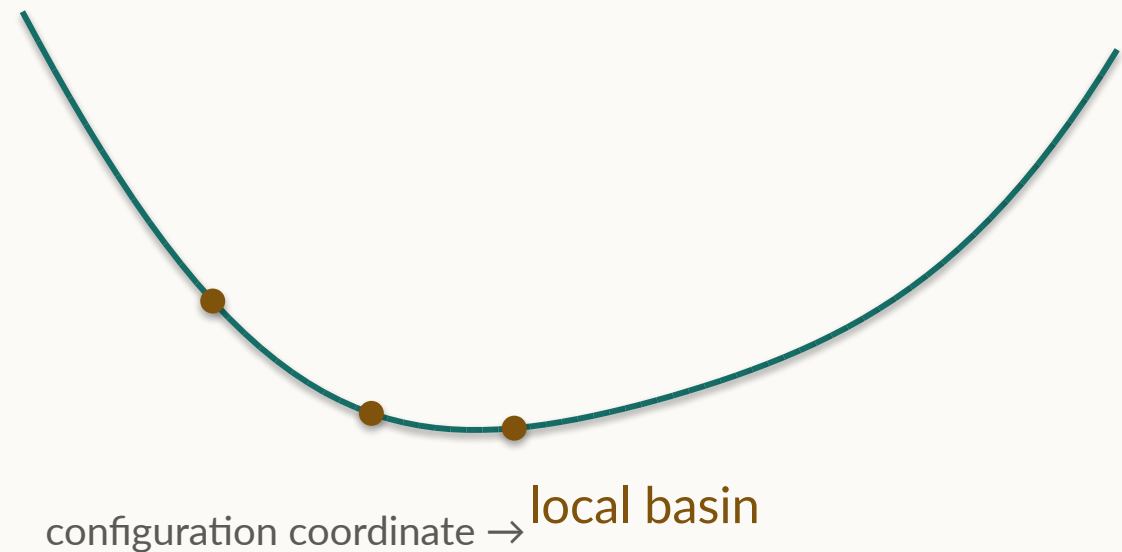


ANALYTICAL 12-6 PAIR · REDUCED UNITS

Minimization searches a local basin

Move toward a configuration with smaller residual forces.

- Start from a structure and a fixed Hamiltonian.
- Choose convergence tolerances and movable degrees of freedom.
- A local minimum is neither a thermal ensemble nor a proof of truth.



SCHEMATIC ENERGY LANDSCAPE · NO MATERIAL DATA

MD follows a time evolution

Mass and force determine acceleration; an ensemble defines state.

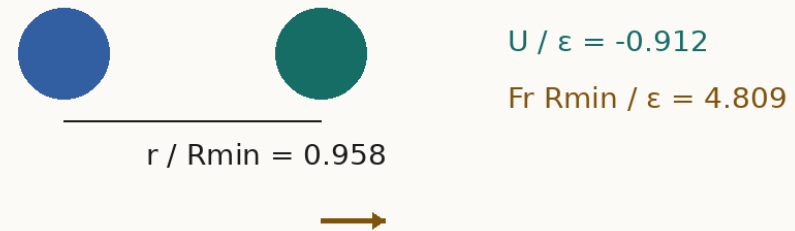
Newton: $m_i d^2 r_i / dt^2 = F_i$.

Energy → force → operation

$$U = \epsilon [(R_{\min}/r)^{12} - 2(R_{\min}/r)^6]$$

Time step, thermostat and barostat require explicit choices.

Equilibrate, then sample; inspect drift and correlations.



Minimize: seek a lower-energy configuration

MD: integrate forces with mass, time step and ensemble

Illustrative prescribed two-atom motion; not an IFF materials trajectory

PLAYABLE TWO-ATOM ILLUSTRATION · NOT MATERIALS MD

An observable needs a defined estimator

A trajectory is not yet the quantity you will compare.

Density

$$\rho = \text{total mass} / \text{volume}$$

NPT state and sampling window

Structure

$g(r)$, coordination, lattice spacing

normalization and phase definition

Mechanical response

$$K = -V(\partial P / \partial V)_T$$

isothermal / adiabatic distinction

Transport

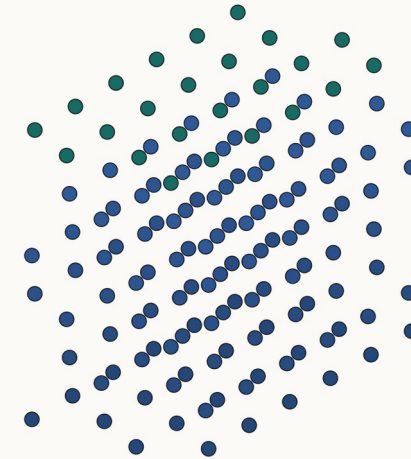
$$D = \lim \text{MSD}(t) / (6t)$$

diffusive regime and finite-size effects

Cleave, then normalize the energy

Two equivalent new surfaces introduce a factor of two.

- | $\gamma \approx (E_{\text{cleaved}} - E_{\text{unified}})/(2A)$.
- | Compare systems with consistent atom counts and cell geometry.
- | This energy estimator approximates free energy only under stated assumptions.



$$\gamma \approx \Delta E / 2A$$

Match the experimental state

A comparison is meaningful only when definitions align.

Material

composition · crystal phase · purity · surface

State

temperature · pressure · environment

Measurement

observable definition · protocol · uncertainty

Simulation

ensemble · finite size · convergence ·
averaging

FITTED, MEASURED AND PREDICTED ARE DIFFERENT LABELS

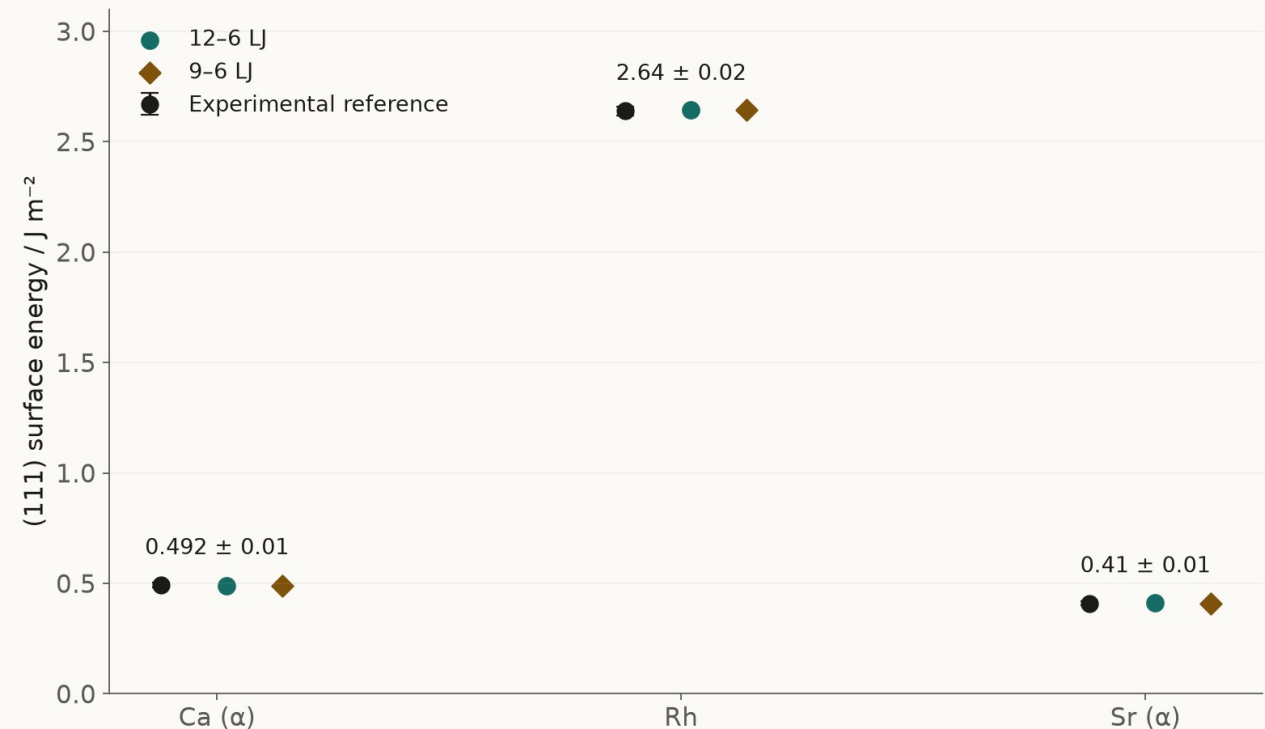
Published fit: Rh matches its targets

Agreement on calibration quantities follows the fitting objective.

5-cell lattice: $19.016 \rightarrow 19.016 \text{ \AA}$ (12-6); $19.014 \text{ \AA}$ (9-6).

Rh γ_{111} reference: $2.64 \pm 0.02 \text{ J/m}^2$.

Both published Rh models: $\gamma_{111} = 2.643 \text{ J/m}^2$.



TABLES 2-3 · 298 K · CALIBRATION, NOT INDEPENDENT VALIDATION

Independent predictions reveal the limits

A good surface fit can coexist with a poor elastic prediction.

- Rh K: experiment 276 GPa; 12-6 LJ 258; 9-6 LJ 175.
- Relative deviations: -6.5% and -36.6%.
- The pair-model elastic constraint limits what can be matched.

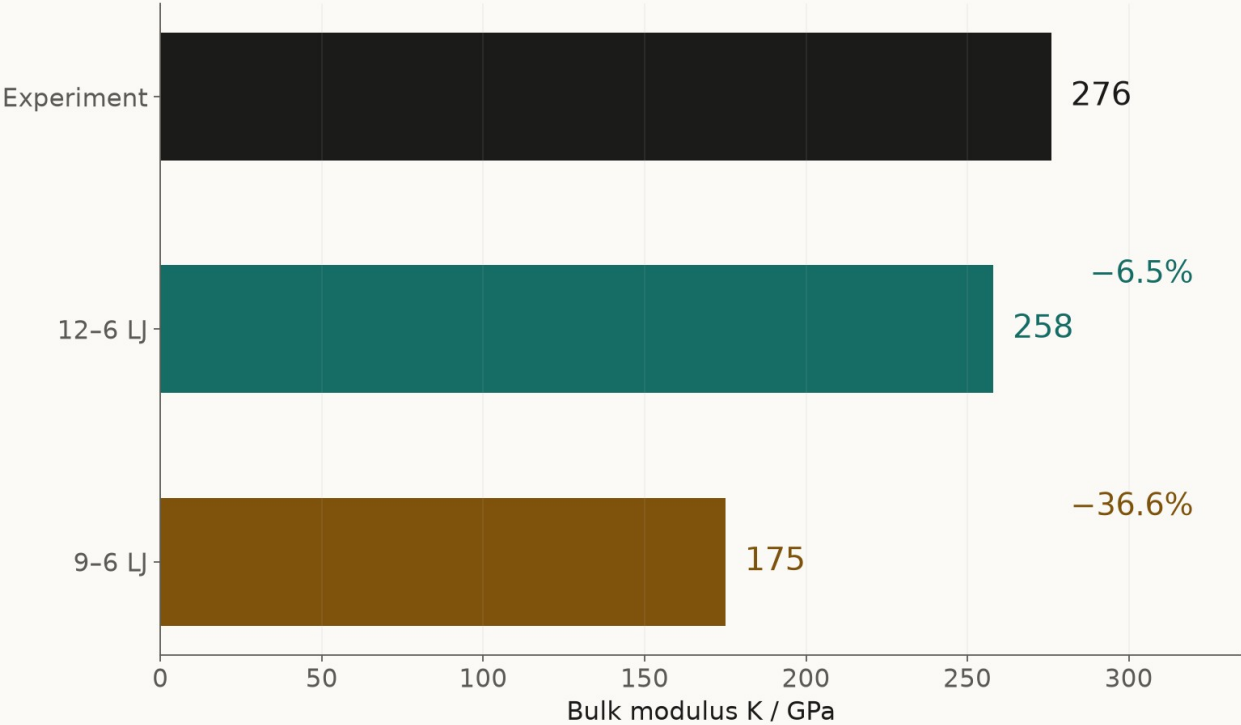
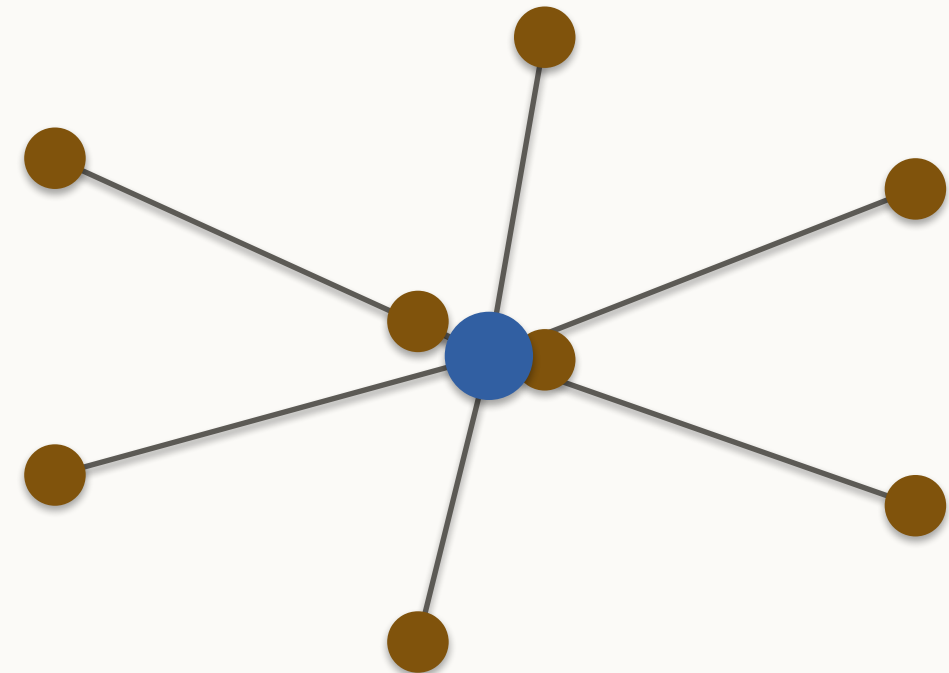


TABLE 5 · NON-FITTED K · SI PROTOCOL REPEATABILITY $\approx \pm 3\%$

Alloys can add chemical polarity

Charge assignment follows chemistry, not the element name alone.

- Liu SI: match Al–Ni alloy-formation evidence to model charges.
- AlNi example: base charges $\pm 0.39e$ in SI Table S2.
- Charge transfer must preserve total neutrality.



Al $+0.39e$

Ni $-0.39e$

B2 topology · schematic · not a metric structure

BINARY ALLOY EXAMPLE · NO GENERAL HEA CLAIM

A defect changes its local environment

Different charge hypotheses change the computed energy.

■ Ni vacancy in AlNi: redistribute the removed site's charge.

■ SI Table S2 raw energies: 5.49, 4.57 and 2.59 eV.

■ Raw defect energies are not final formation free energies.

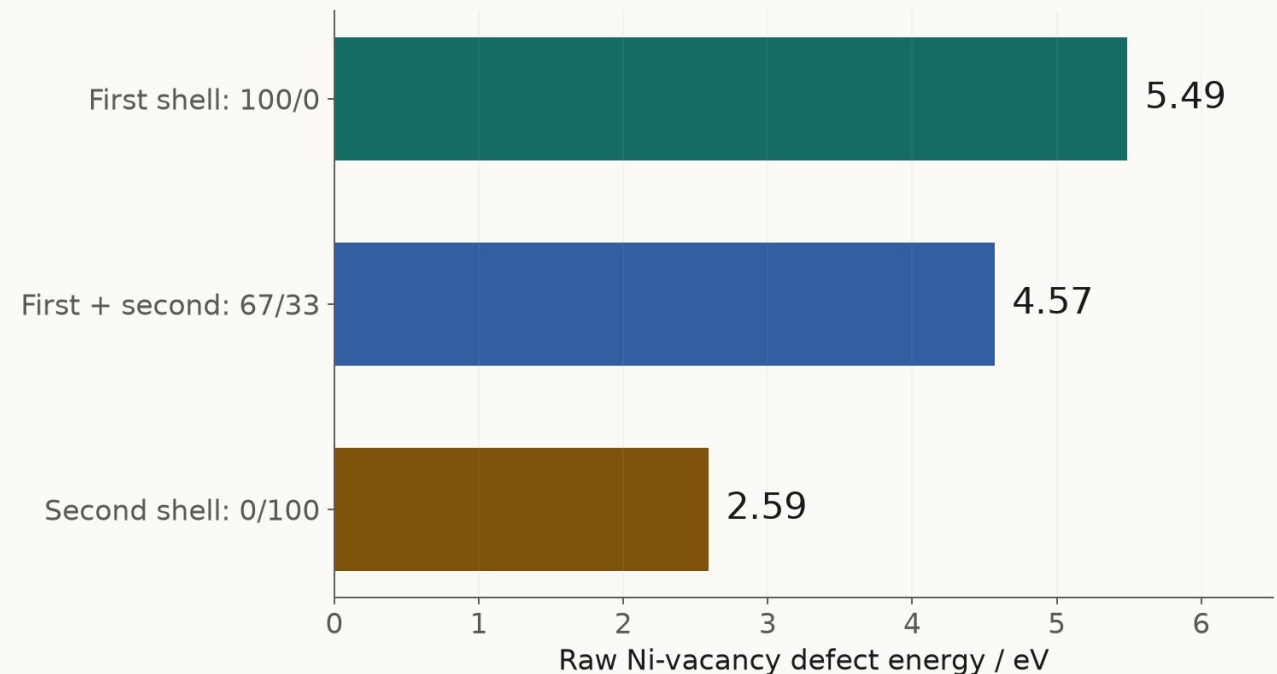


TABLE S2 · CHARGE-HYPOTHESIS SENSITIVITY

Calibration and validation take different paths

Protect the independent test from every tuning decision.

Calibration

references → fit parameters → verify
targets

Independent test

reserved property → forward prediction
→ compare



qualified for scope · limited · withheld

A revised model needs fresh evidence

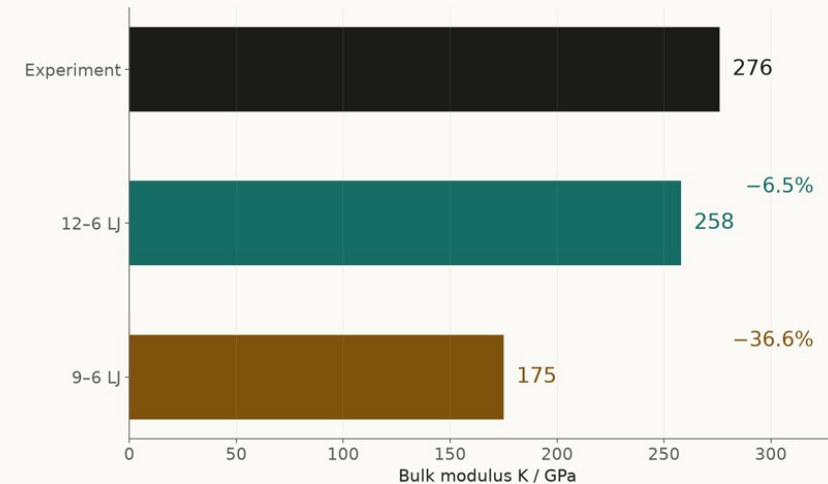
Once a test informs a revision, it is no longer untouched.

- Changing parameters or model family creates a new candidate.
- Retire reused test data into development evidence.
- Reserve fresh observations for the next independent claim.

Calibration → independent prediction

Test: bulk modulus was not fitted

$R_h \cdot K$

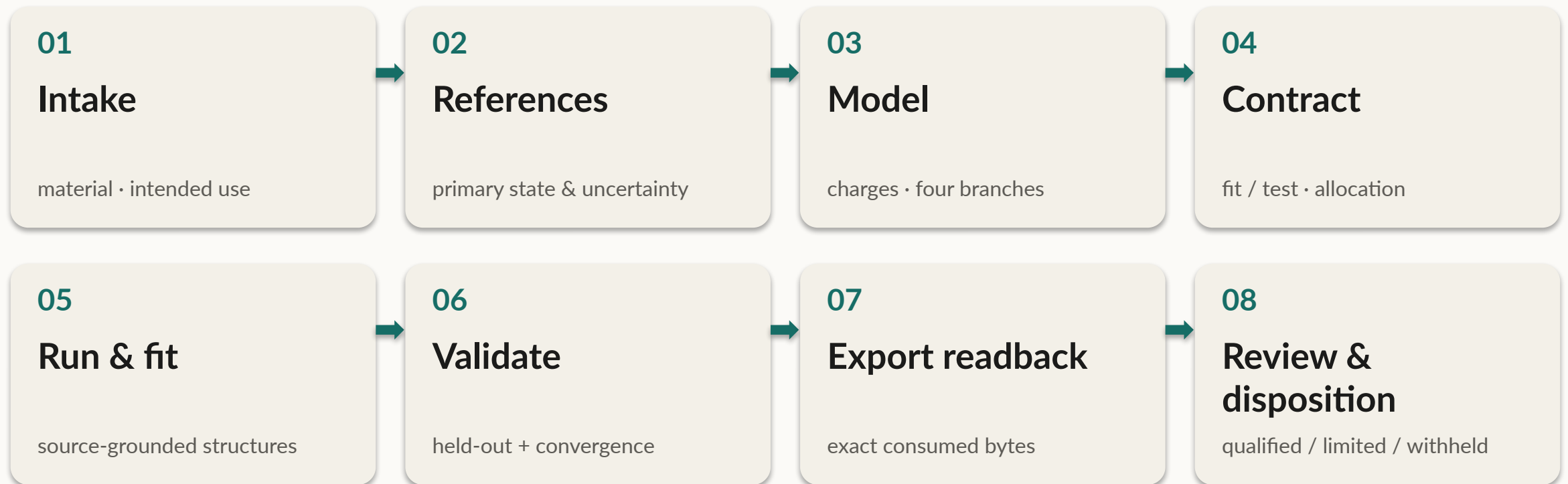


Published Kanhaiya 2021 Tables 2, 3, 5; animation reveals data, not new simulation

EVIDENCE HYGIENE · NO NEW FIT OR VALIDATED REVISION CLAIM

The IFF Agent carries the evidence chain

The workflow joins scientific choices to inspectable files.



IMPLEMENTED EVIDENCE WORKFLOW · SCIENTIFIC ACCEPTANCE IS SEPARATE

Scientific decisions define the scope

References, model choices, test rules and final adoption stay explicit.

References

primary experiments · state · derivation

Model

physical charges · family · neutrality

Fit / test rules

variables · bounds · validation · resources

Scientific adoption

exact reviewed export · supported scope

IMPLEMENTED WORKFLOW · CONSEQUENTIAL SCIENTIFIC DECISIONS

Inputs become a reproducible dossier

Each output should point back to the input and model revision.

Inputs

composition / phase / state; intended observable; primary references; structural origin

Outputs

branch-specific coefficients; structures and runnable inputs; readback comparisons; predictions, uncertainty and limits

ILLUSTRATIVE INPUT / OUTPUT CONTRACT · NOT A COMPLETED EXPORT

Acceptance is more than execution

A runnable package still needs independent scientific evidence.

Execution

A completed run establishes a result under its stated inputs.

Scientific checks

Match fit targets, test reserved properties and verify convergence.

Disposition

Adopt a qualified or limited scope, or withhold the package.

SCIENTIFIC ACCEPTANCE REQUIREMENTS

Choose the method for the decision

Compare specific models, chemistry and observables.

Classical / IFF

REPRESENTATION

Defined potential and conventions

USEFUL ROLE

Large-system sampling, interfaces

WHAT TO CHECK

Applicability and missing physics

DFT

REPRESENTATION

Electronic-structure approximation

USEFUL ROLE

Bonding, electronic effects, reference calculations

WHAT TO CHECK

Functional, state and convergence

MLIP

REPRESENTATION

Learned energy/force representation

USEFUL ROLE

Rich local environments and efficient sampling

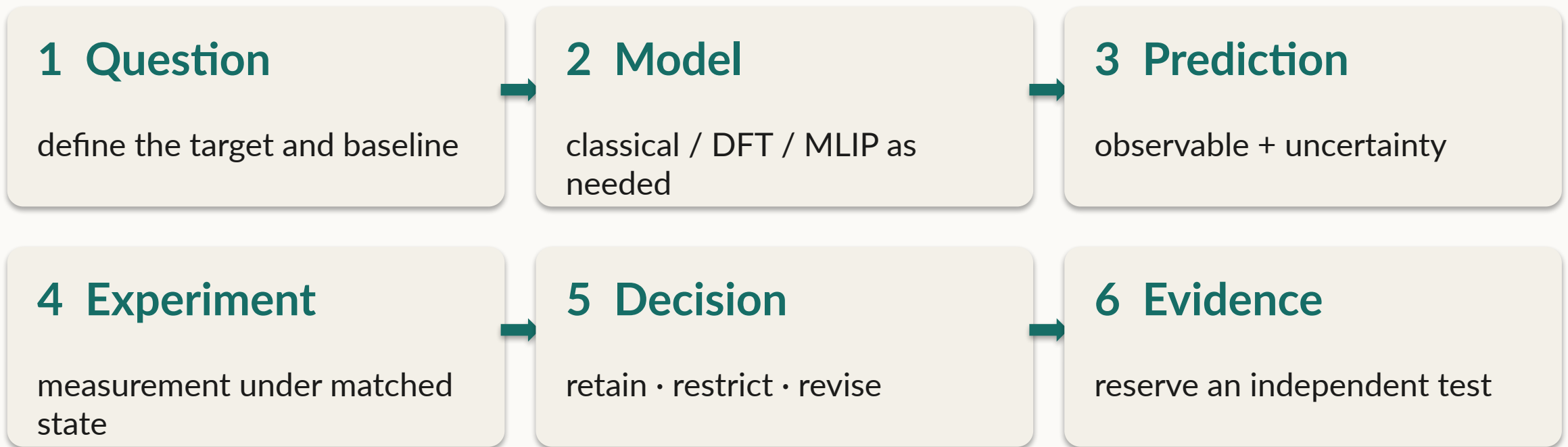
WHAT TO CHECK

Training coverage and extrapolation

COMPLEMENTARY ROLES • NO UNIVERSAL SPEED / ACCURACY RANKING

Connect methods to an experimental loop

Use each calculation to change the next physical decision.



Make the first question narrow and testable

A useful model comes with a defined boundary of trust.

Inputs

material / phase / interface; decision and observable; primary data and uncertainty; source structures

Outputs

an exact model package; reproducible predictions; fit-versus-test evidence; limitations and failure conditions

ATOMS → MODEL → OBSERVABLE → EVIDENCE → DECISION

Appendix • equations and conventions

Always name the functional form before copying coefficients.

12-6 • Rmin convention

$$U = \epsilon[(R_{\min}/r)^{12} - 2(R_{\min}/r)^6]$$

9-6 • Rmin convention

$$U = \epsilon[2(R_{\min}/r)^9 - 3(R_{\min}/r)^6]$$

Electrostatics

$$U_{ij} = q_i q_j / (4\pi\epsilon_0 \epsilon_r r_{ij}), \text{ with a defined long-range treatment}$$

Forces and dynamics

$$F_i = -\nabla_{r_i} U; m_i \ddot{r}_i = F_i$$

Appendix • the published data remain visible

Selected reference entries; do not hide failed predictions.

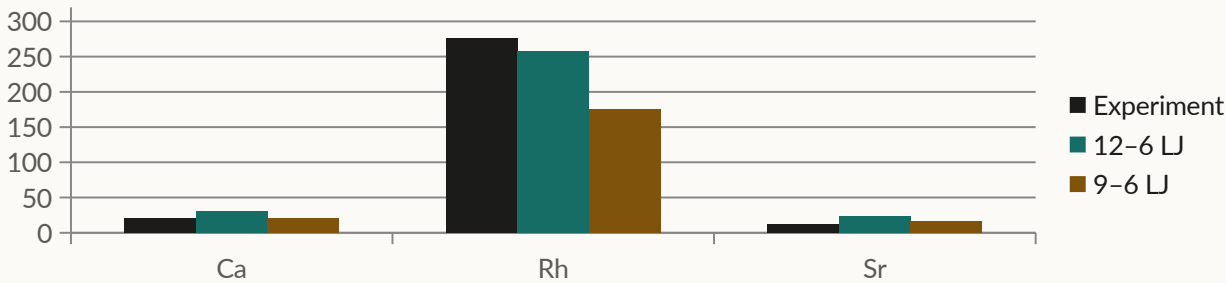
Metal	5a expt / Å	5a 12-6 / Å	γ_{111} expt / J m ⁻²	γ_{111} 12-6	K expt / GPa	K 12-6 / GPa
Ca (α)	27.942	27.947	0.492 $\pm$ 0.01	0.490	20	30
Rh	19.016	19.016	2.64 $\pm$ 0.02	2.643	276	258
Sr (α)	30.420	30.423	0.41 $\pm$ 0.01	0.411	12.0	24

Not fitted: bulk modulus K / GPa

K references are selected, not averaged.

SI S7-S8: repeatability about $\pm 3\%$.

Protocol agreement is separate from accuracy.



Appendix • what uncertainty must include

SI mechanical repeatability is approximately $\pm 3\%$, separate from accuracy.

Sampling

autocorrelation • independent blocks •
equilibration

Numerical

step / cutoff / size / slab / solver
convergence

Experimental

state • protocol • specimen • measurement
error

Model

chemical scope • missing physics •
extrapolation

Appendix • primary references

Source details and changes are retained in the companion manifest.

Kanhaiya et al. (2021) • FCC-metal parameters, methods and results

Author Correction (2021) • restored structure and script data

Liu et al. (2018) • alloy bonding (primary paper citation)

Liu Supporting Information • S15–S19, Tables S1–S2

LAMMPS • lj/cut parameter convention

LAMMPS • velocity Verlet / minimization

Batatia et al. (2022) • MACE (retrieved v2, 2023)

IFF Agent workflow documentation (2026)

Appendix • reuse and scientific limits

Literature evidence, original visuals, and explicit limitations.

Original assets

Geometry renders, plots, diagrams and silent media; CSV data and production scripts. No additional reuse license is granted.

Scientific limits

No new materials MD, parameter fit or adopted package; binary-alloy scope; source-derived geometry and illustrative motion.